MIL-B-27044C(USAF)
23 February 1968
SUPERSEDING
MIL-B-27044B(USAF)
13 May 1963

MILITARY SPECIFICATION

BOMB, PRACTICE BDU-12/B

1. SCOPE

1.1 This specification covers the requirements for one type of practice bomb designated as the BDU-12/B, hereafter referred to as the practice bomb. The practice bomb is designed to simulate cartridge ejected, parachute retarded delivery provisions of the parent weapon.

2. APPLICABLE DOCUMENTS

* 2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

L-P-378	Plastic Sheet And Strip (Polyolefin)
PPP-B-636	Box, Fiberboard

Military

MIL-P-116	Preservation, Methods Of
MIL-D-1000	Drawings, Engineering And Associated Lists
MIL-E-5272	Environmental Testing, Aeronautical And Associated Equipment, General Specification For
MIL-B-26195	Boxes, Wood-Cleated, Skidded, Load-Bearing Base
MIL-W-27000	Weapon, Nuclear, Practice And Training-And Similar Special Weapons Items, General Specification For
MIL-C-27556	Cartridge Kit, Parachute Ejection MLU-14/B And Cartridge Kit, Parachute Ejection MLU-15/B

STANDARDS

Military

MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of US Military Property

FSC 1105

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MIL-STD-794	Parts And Equipment, Procedures For Packaging And Packing Of
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation Of
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, And Waterproofing; With Appropriate Test Methods

DRAWINGS

Air Force

61J14050	Bomb, Practice BDU-12/B
61J14051	Case Section Assembly
61J14052	Aft Section Assembly
61J14054	Nose Section Assembly
61D14059	Cap, Parachute, Aft End
61B14061	Pin, Straight, Headless
61D14063	Ring, Suspension Line, Front
61C14118	Tube, Outside
61D14986	Cartridge Kit, Parachute Ejection MLU-14/B
61D14987	Cartridge Kit, Parachute Ejection MLU-15/B
62B4307	Parachute
62A13072	Parts Kit, Practice Bomb KMU-163/C
63C13046-3	Parachute Assembly
64D13111	Clamp, Aft

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Official Classification Committee

Uniform Freight Classification Rules

(Application for copies should be addressed to the Official Classification Committee, Room 202, Union Station, 516 West Jackson Boulevard, Chicago, Ill. 60606.)

3. REQUIREMENTS

* 3.1 General specification. The requirements of MIL-W-27000 shall apply as requirement of this specification with the exceptions and additions specified herein. In case of conflict between the two specifications, this specification shall govern.

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3.2 Preproduction testing. This specification makes provision for preproduction testing.

* 3.3 Components. The practice bomb shall consist of the following major components:

ITEM	PART NUMBER	NOMENCLATURE	QUANTITY
a	61J14051	Case Section Assembly	1
b	61J14052	Aft Section Assembly	1
c	61J14054	Nose Section Assembly	1
d	61D14986	Cartridge Kit, Parachute Ejection MLU-14/B	1
	or		
	61D14987	Cartridge Kit, Parachute Ejection MLU-15/B	1
e	62B4307	Parachute	1
f	64D13111	Clamp, Aft	1

3.4 Design and construction. The practice bomb and components shall be designed and constructed in conformance to drawing 61J14050 and subsidiary drawings.

3.4.1 Shear force. The load required to shear the shear joint shall be 6600 to 7200 pounds.

3.4.2 Interchangeability. Parts functionally and dimensionally interchangeable shall be identified by the same part number. When parts are not interchangeable, the criteria listed in MIL-D-1000 shall apply and new drawings shall be prepared as necessary.

3.4.3 Reusability. The practice bomb, after being dropped, shall be reusable when reassembled with replacement parts from the 62A13072 Parts Kit, Practice Bomb KMU-163/C.

* 3.4.4 Weight. The weight of the practice bomb shall be 500 pounds ± 15 pounds.

* 3.4.5 Center of gravity. Center of gravity of the practice bomb shall be station 54.60 ± 1 inch.

3.4.6 Location reference. Station 0 shall be the forward surface of the nose of the practice bomb.

3.4.7 Pitch and yaw moment of inertia. Pitch and yaw moment of inertia of the practice bomb shall be 430000 lb-in^2, plus 10 percent minus zero.

3.5 Environmental requirements. The practice bomb shall be capable of meeting the performance requirements of this specification during or after, as applicable, exposure to the following environmental conditions:

a. High temperature to 71° Centigrade (C) (160° Fahrenheit (F)).

b. Low temperature to -54°C (-65°F).

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c. Temperature shock incident to service use.

d. Salt spray likely to be encountered in storage or transportation near a large body of salt water.

e. Shock likely to be encountered during transportation or handling in service use.

f. Vibration incident to transportation and flight in carrier aircraft.

* 3.6 Finish. Finish requirements for the practice bomb shall be as specified in MIL-W-27000.

3.6.1 Final finish coat. Final finish coat shall be as specified in the engineering drawings.

3.7 Identification of product. The practice bomb and all parts and assemblies shall be marked in accordance with MIL-STD-130.

* 3.8 Government furnished property. Unless otherwise specified in the contract, the following item shall be furnished to the contractor, upon his request, as Government furnished property for the purpose of fabrication of the practice bomb:

PART NUMBER	NOMENCLATURE	QUANTITY
62B4307	Parachute	1 per practice bomb (see 6.2)

* 3.9 Government loaned property. When specified in the contract, the Government will loan the following for use in testing the practice bomb to the contractor upon his request.

ITEM NUMBER	PART NUMBER	ITEM DESCRIPTION	QUANTITY
1	61D14986	Cartridge Kit, Parachute Ejection MLU-14/B (Inert)	2
or			
1	61D14987	Cartridge Kit, Parachute Ejection MLU-15/B (Inert)	2
2	62B4307-2	Dummy Parachute	2

* 3.10 Workmanship. Workmanship shall be as specified in MIL-W-27000.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein,

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unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. Inspection and testing of the practice bomb shall be classified as follows:

a. Preproduction tests

b. Quality conformance tests

4.3 Test conditions. Test conditions shall be as required for each specific test.

4.4 Preproduction testing.

4.4.1 Test sample. The preproduction test units shall be representative of the production equipment. They shall be tested at a laboratory designated by the procuring activity, or, when so stated in the contract, at the contractor's facility under the supervision of the procuring activity.

* 4.4.2 Test report. In the event that preproduction testing is performed at the contractor's plant, the contractor shall, after completion of the preproduction tests, prepare a preproduction test report in accordance with MIL-STD-831 and furnish three complete copies of the report to the procuring activity.

4.4.3 Preproduction tests. Preproduction tests shall consist of all tests described in 4.6.

4.5 Quality conformance tests. Quality conformance tests shall consist of the following individual tests:

a. Examination of product

b. Conformance to drawings

4.6 Test methods. All tests shall be performed with the practice bomb completely assembled.

NOTE: In no case shall live explosive devices be assembled in the bomb being tested. Inert models only shall be used.

* 4.6.1 Examination of product. Each practice bomb shall be examined for strict conformance to the requirements of this specification. Failure to meet these requirements shall be cause for rejection.

4.6.2 Conformance to drawings. Each practice bomb shall be inspected for conformance to the drawings and specifications and standards specified therein. Failure to conform to these criteria shall be cause for rejection.

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4.6.3 Static load tests.

* 4.6.3.1 Suspension tests. Suspension tests shall be performed once simulating 30 inch suspension system provisions and once simulating 14 inch suspension system provisions. The suspension tests shall consist of the following:

a. Procedure I of the suspension test specified in MIL-W-27000.

b. Procedure II of the suspension test of MIL-W-27000.

EXCEPTION: Longitudinal loads shall not be applied.

Design limit loads shall be:

Up	4165 pounds
Down	3675 pounds
Lateral	980 pounds

Rejection criteria of MIL-W-27000 shall apply.

4.6.3.2 Lug tests.

4.6.3.2.1 Test set up. The representative bomb shall be secured to a test stand by straps of sufficient width to prevent compression damage.

* 4.6.3.2.2 Test procedures. Tensile loads, applied by any convenient means shall be applied to the lugs in the following manner:

a. Forward 30 inch suspension (folding) lug.

(1) 16134 pounds at an angle of 1.2 degrees right of vertical. Physical damage or permanent deformation shall be cause for rejection.

(2) 7260 pounds at an angle of 66 degrees right of vertical. Physical breaking or tearing out of the lug shall be cause for rejection. Permanent deformation at this load is permissible.

b. Forward 14 inch suspension (threaded) lug.

(1) 8505 pounds at an angle of 58 degrees aft of vertical. Physical damage or permanent deformation shall be cause for rejection.

(2) 14120 pounds at an angle of 7 degrees aft at vertical. Physical failure or tearing out of the lug shall be cause for rejection. Permanent deformation at this load is permissible.

c. Aft lug (30 and 14 inch suspension).

(1) 14850 pounds at an angle of 27 degrees aft of vertical. Physical damage or permanent deformation of the lug shall be cause for rejection.

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(2) 21900 pounds at an angle of 9 degrees aft of vertical. Breaking or tearing out of the lug shall be cause for rejection. Permanent deformation at this load is permissible.

4.6.3.3 Tail fin tests.

* 4.6.3.3.1 Test set up. The test set up shall be in accordance with MIL-W-27000.

4.6.3.3.2 Test procedures.

* 4.6.3.3.2.1 Load application. Concentrated loads shall be applied at station 103.36, 3.28 inches outboard of the intersection of the bomb skin and tail fin in the manner specified in MIL-W-27000.

* 4.6.3.3.2.2 Sequence and intensity. The loads shall be applied in the following steps.

a. Preload the fins to 720 pounds and establish a zero reference for deflection.

b. Increase the load on each fin to 1080 pounds.

c. Increase the loads on each fin in increments of 110 pounds to 2300 pounds, decreasing load to 720 pounds after each increment. Measure fin deflection at each load increment up to and including total load. Breaking or permanent deformation in excess of 0.10 inch at the outboard edge at station 106.00 at any load increment shall be cause for rejection.

d. Reduce loads to 720 pounds.

e. Increase the loads to 2300 pounds.

f. Increase the loads in increments of 56 pounds to 3000 pounds.

g. Reduce the loads simultaneously to 720 pounds. Failure during or after loading to 3000 pounds shall be cause for rejection. Permanent deformation is permissible.

4.6.4 Shear joint tests.

4.6.4.1 Test set up.

a. The aft section assembly shall be rigidly assembled to a fixture which will withstand the loads to be applied.

b. The tail cap shall be assembled to the aft section assembly by use of the 6 shear pins.

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4.6.4.2 Procedures. Loads shall be applied to the tail cap in the following manner:

a. 6600 pounds aft along longitudinal axis. The shear pins shall retain the tail cap.

b. 7200 pounds aft along the longitudinal axis. The shear pins shall shear and release the tail cap.

Failure to meet the above requirements sh ll be cause for rejection.

* 4.6.5 Weight test. The practice bomb shall be weighed using a scale calibrated to a standard traceable to the National Bureau of Standards. The weight shall be 500 pounds ± 15 pounds. Weight not falling within these tolerances shall be cause for rejection.

* 4.6.6 Center of gravity. The center of gravity of the practice bomb shall be determined by a convenient means. The center of gravity shall be at station 54.60 ± one inch. A determined center of gravity not within these tolerances shall be cause for rejection.

4.6.7 Moments of inertia test.

* 4.6.7.1 Fixture and preparation. The fixture for this test shall be a simple torsion pendulum or bar with an adjustable hanger which can be attached to the lugs of the practice bomb. This fixture shall be calibrated by swinging a bar of known weight and calculated moment of inertia. The periods (in seconds/oscillation) obtained from this swinging shall, by substitution in the following formula, be used to determine the spring constant of the torsion bar:

$$I_1 = C_Z \left(\frac{P}{2\pi}\right)^2 - I_2$$

Where I_1 = measured inertia (slug ft^2)

C_Z = spring constant of torsion bar (ft-lb/radian)

P = period (seconds/oscillation)(time of one complete cycle)

I_2 = moment of inertia of hanger (slug ft^2)

4.6.7.2 Test procedure. The practice bomb shall be attached to the hanger and swung, and the formula in 4.5.7.1 shall be used to determine moment of inertia of the bomb. Pitch and yaw moment of inertia shall be 430000 lb-in^2, plus 10 percent, minus zero. Determined moment of inertia not within these tolerances shall be cause for rejection.

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* 4.6.8 Environmental tests. The practice bomb shall be subjected to the following environmental tests in the sequence in which listed:

TEST	TEST SPECIFICATION	PROCEDURES
a. High Temperature	MIL-E-5272	II
b. Low Temperature	MIL-E-5272	I
c. Temperature Shock	MIL-E-5272	I
d. Salt Spray	MIL-E-5272	I
e. Shock	MIL-B-27000	

f. Vibration test is to be in accordance with method 514.1 of MIL-STD-810, Equipment Class 3, Mounting Method A, Test Curve B.

Permenent deformation or failure of any part shall be cause for rejection.

4.9 Inspection of the preservation, packaging, packing, and marking for shipment and storage. Sample items or packs and the inspection of the preservation, packaging, packing and marking for shipment and storage shall be in accordance with the requirements of Section 5, or the documents specified therein.

* 5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Levels A and B. The fin mounting screws and fins shall be removed from the BDU-12/B and shall be packaged within fiberboard boxes conforming to PPP-B-636, Type CF or SF, class weather resistant, Grade V3, which shall be secured in a suitable manner within the bomb shipping container. Each BDU-12/B shall be preserved in accordance with MIL-P-116, Method III. After cleaning (any suitable MIL-P-116 cleaning procedure), each bomb shall be shrouded with plastic film. The film shall be opaque and shall conform to L-P-378, type I or II, grade B or equal. Minimum thickness of the film shall be 0.0060 inch. The shroud shall be one solid piece and shall be draped over the bomb in such a manner as to prevent entry of rain or foreign matter from the top, sides or ends of the bomb when it is in the crated position. To prevent a water pocket in event of film rupture, the film shall not overlap on the bottom of the crated bomb. The ends of the film shall be gathered, folded and taped or tied in a manner that will prevent entry of rain during shipment. To retain the shroud in place, it shall also be tied with a suitable cord or fastened with tape in at least two points along the body of the bomb. If tape is used, the tape shall not come in direct contact with the painted surface of the bomb.

5.1.2 Level C. Cleaning, drying, preservation and packaging shall be in accordance with manufacturer's commercial practice.

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5.2 Packing.

5.2.1 Levels A and B. Each BDU-12/B preserved to meet 5.1.1 shall be packed in a box conforming to MIL-B-26195, class 2, style A, type I. Cushioning, blocking and bracing shall be in accordance with MIL-STD-1186 or as specified by the procuring activity.

5.2.2 Level C. Each BDU-12/B preserved to meet 5.1.2 shall be packed in a manner to insure carrier acceptance and safe delivery to destination. Container shall be in accordance with Uniform Freight Classification Rules or regulation of other carriers applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, the shipping container shall be marked in accordance with the requirements of MIL-STD-129. The nomenclature shall be: Bomb, Practice BDU-12/B.

6. NOTES

6.1 Intended use. The BDU-12/B Practice Bomb is intended for use in training aircrews in delivery of the retarded version of the parent weapon. The practice bomb with application of the KMU-163/C Parts Kit is reusable.

* 6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Number of KMU-163/C Parts Kit, Practice Bomb to be furnished with each order.

NOTE: This item may be ordered separately.

c. Provisions for making the following items available to the Government parachute packing activity for assembly of the parachute in conformance to drawing 63C13046-3:

	ITEM	DRAWING NUMBER	QUANTITY PER PARACHUTE
1.	Cap, Parachute, Aft End	61D14059	1
2.	Ring, Suspension Line, Front	61D14063	1
3.	Tube, Outside	61C14118	1
4.	Pin, Straight, Headless	61B14061	4
5.	Internal Wrenching Screw (Nyloc Company Division 611 Industrial Avenue Paramus, NJ 07652)	NK520-10R10 (or equal)	4

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d. Whether preproduction tests will be conducted by the procuring activity.

e. Conditions under which Government loaned property will be made available to the contractor.

f. Levels of preservation, packaging and packing.

6.2.1 Cartridge kits. The MLU-14/B or MLU-15/B Cartridge Kits should be procured separately.

6.2.2 Preproduction test samples. It is expected that the contract or purchase order will specify that a minimum of two units representative of the production practice bomb will be submitted to the procuring activity for test and evaluation. The invitation for bids, contract or purchase order should specify the point of inspection for these tests.

6.3 Production delivery. No production practice bomb should be delivered prior to full approval of the preproduction samples.

6.4 Definitions.

a. Preproduction tests. Preproduction tests are those tests performed prior to production on representative samples of the production equipment to demonstrate that the equipment meets the requirements of this specification.

b. Quality conformance tests. Quality conformance tests are those tests accomplished on representative samples of equipment submitted for acceptance under a contract to determine conformance to the requirements of this specification, and to determine that the equipment is equivalent in performance and construction to the approved preproduction samples.

* 6.5 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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